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Forecasting U.S. New-Home Months' Supply: A Multivariate SARIMAX Analysis of Housing Starts and Mortgage Rates

Matilda Kumi, Department of Mathematics and Statistics, East Tennessee State University; Dr. Mostafa Zahed, Department of Mathematics and Statistics, East Tennessee State University

ABSTRACT

In the United States, the availability of housing has become a growing concern among households, since fluctuations in new-home months' supply affect affordability and market stability. This study examines the Monthly Supply of New Houses (MSACSRNSA), a not-seasonally-adjusted months'-supply ratio, using a two-stage Box-Jenkins approach. A univariate analysis of 757 monthly observations, January 1963 to January 2026, identifies a SARIMA(1,1,0)×$(1,1,1)_{12}$ model without a constant term as the most robust order; refit on the common 1971–2026 sample used for later comparison, this specification achieves an AIC of 1303.46. A twelve-month holdout evaluation of this exact model, refit on the training data, produced MAE = 0.628, RMSE = 0.717, and MAPE = 7.02%.

The baseline model was then extended to a multivariate SARIMAX framework, to test whether external economic conditions improve model fit. Two exogenous variables were introduced: New Privately-Owned Housing Units Started (HOUST), a seasonally adjusted annual rate, and the 30-Year Fixed Rate Mortgage Average (MORTGAGE30US). The common sample across all three series spans April 1971 through January 2026: 658 monthly observations before differencing, and 645 effective observations after the regular and seasonal differencing used in estimation. Both regressors were first-differenced; the HOUST level series showed strong persistence, while ADF testing confirmed stationarity after first differencing for both HOUST and MORTGAGE30US. Each predictor was tested individually first, then combined into one joint SARIMAX model.

Differenced housing starts significantly improved model fit on their own ($p = 0.0051$, AIC = 1297.98). Differenced mortgage rates did better on their own ($p < 0.0001$, AIC = 1288.13, SBC = 1306.00). Combining both predictors achieved the lowest AIC of any model tested (1286.59), though the mortgage-only model retained the lowest SBC (1306.00 versus 1308.94), so the two criteria disagree on whether the added HOUST parameter earns its keep. In the joint model, the mortgage-rate coefficient stayed significant ($p = 0.0002$), while the housing-starts coefficient dropped to marginal significance ($p = 0.0569$), consistent with a modest negative correlation between the two differenced regressors ($r = -0.130$, $p = 0.0008$). A conditional twelve-month holdout comparison was also conducted, using the realized (actual, historical) HOUST and MORTGAGE30US values for the holdout months. Because these exogenous values were not known in advance, this exercise does not represent a true ex-ante forecast. All three SARIMAX specifications nonetheless produced lower MAE, RMSE, and MAPE than a seasonal-naive benchmark; against the univariate baseline (MAE = 0.628, RMSE = 0.717, MAPE = 7.02%), the HOUST-only model improved on every metric (MAE = 0.614, RMSE = 0.705, MAPE = 6.91%), while the mortgage-only and joint models improved MAE and MAPE but not RMSE. Mortgage rates carry most of the incremental information beyond the univariate baseline in-sample, but housing starts produced the most consistent improvement across the three error metrics in this conditional holdout window.

Keywords: Time Series Forecasting, SARIMAX Modeling, New-Home Months' Supply, Box-Jenkins Methodology, Macroeconomic Analysis

INTRODUCTION

Housing availability is a persistent concern for households and policymakers in the United States. When the supply of new homes becomes constrained relative to demand, it can place upward pressure on prices and make it harder for households to secure housing. This paper studies fluctuations in the Monthly Supply of New Houses in the United States (MSACSRNSA) using a two-stage Box-Jenkins approach: a univariate seasonal time series model is first established as a baseline, and this baseline is then extended to incorporate economic conditions outside the housing data itself.

Prior research shows that housing-supply responsiveness shapes how housing markets adjust to demand and price changes (Glaeser, Gyourko, and Saiz 2008), while financing conditions affect housing demand and broader housing-market activity, including construction and permits (Fuster and Zafar 2021; Časta 2025). This paper examines whether two such factors, new housing starts and 30-year mortgage rates, carry information about new-home months' supply beyond what the series' own seasonal history already provides.

In the first stage, a univariate baseline is developed from 757 monthly observations spanning January 1963 to January 2026. The model order is selected using information criteria and residual diagnostics, and validated against a twelve-month holdout period. In the second stage, that baseline is extended into a Seasonal Autoregressive Integrated Moving Average with Exogenous Variables (SARIMAX) framework, testing whether new housing starts (HOUST) and the 30-year fixed-rate mortgage average (MORTGAGE30US) improve on the univariate model, first individually and then jointly.

Because months' supply combines an inventory level with a sales rate, it can shift noticeably from one month to the next as either component changes. Housing starts and mortgage rates are examined here as external economic covariates that may carry information about these changes. The analysis does not estimate a formal conditional-variance model or attempt structural identification of economic shocks.

The Background section that follows defines months' supply and reviews related research on construction activity and financing conditions; the Methodology section then develops the univariate baseline; the SARIMAX Extension section introduces and evaluates the exogenous predictors; and the Conclusion summarizes the findings and their limitations.

BACKGROUND

The Monthly Supply of New Houses (MSACSRNSA) is not a count of homes; it is the ratio of new houses for sale to new houses sold, expressed as the number of months it would take to sell the current new-home inventory at the prevailing sales pace (U.S. Census Bureau and U.S. Department of Housing and Urban Development, n.d.). Because it combines an inventory level with a sales rate, months' supply can rise either because inventory grows or because sales slow, and distinguishing between these channels is central to interpreting the series. Housing economists generally treat a five- to six-month supply as consistent with a balanced market: readings above that range are associated with softer price growth and construction activity, while readings below it are associated with tighter markets and price pressure (National Association of Home Builders 2024). This paper studies the new-home market specifically, not the much larger stock of existing homes, which is tracked separately and can diverge from new-home supply.

Construction activity is one channel through which new-home supply changes: housing starts initiate new residential construction that can later add to the for-sale inventory once units are completed, with a lag reflecting typical construction timelines. Research on housing-supply responsiveness has emphasized that construction activity, and the elasticity of housing supply to demand and price signals more broadly, varies substantially across markets and shapes how housing markets absorb shocks (Glaeser, Gyourko, and Saiz 2008). Residential construction has also been shown to respond directly to real interest rates and construction-input costs (Blackley 1999), and stock-flow models of housing-market adjustment treat construction as the key channel linking price signals to future inventory (DiPasquale and Wheaton 1994). This literature motivates examining housing starts (HOUST) as a predictor of new-home supply, though it studies metropolitan-level price and construction dynamics generally rather than the specific national months'-supply series analyzed here.

Financing conditions are a second channel. Survey evidence indicates that buyers' willingness to pay for a home responds to the mortgage rate they face, so higher rates can be expected to cool buyer demand at a given price (Fuster and Zafar 2021). Interest rates also affect the supply side directly, since builders finance construction with short-term credit; higher financing costs have been found to slow residential investment through both demand and supply channels (McCarthy and Peach 2002). More recent work has connected mortgage rates specifically, rather than policy rates alone, to house prices and to broader housing activity including construction and permits (Časta 2025), and has shown that the house-price response to monetary policy itself depends on local housing-supply elasticity (Aastveit and Anundsen

2022). Together, this research motivates examining the 30-Year Fixed Rate Mortgage Average (MORTGAGE30US) as a second predictor of new-home months' supply, on the premise that financing conditions influence both how quickly new homes sell and how much new construction gets started. As with the construction-activity literature, this prior work addresses housing demand, prices, and residential investment broadly; it does not directly study the national new-home months'-supply series modeled here.

Despite this research connecting construction activity and financing conditions to housing-market dynamics, relatively little prior research appears to have evaluated these two predictors directly against the U.S. new-home months'-supply series, and their relative and joint contributions to this specific outcome remain underexplored. Prior forecasting work on housing starts themselves, discussed further below, has compared time-series and economic-model approaches, but months' supply, an outcome that reflects both construction and sales together, has not received the same direct attention. This paper addresses that gap by asking whether housing starts and 30-year mortgage rates provide incremental information, individually and jointly, beyond a seasonal univariate model of U.S. new-home months' supply, and if so, which of the two carries more of that incremental information. A two-stage Box-Jenkins approach is used to answer this question: a univariate SARIMA model is first established as a baseline, and this baseline is then extended with each exogenous predictor individually and jointly to isolate their separate and combined contributions.

A Seasonal Autoregressive Integrated Moving Average (SARIMA) framework is well suited to this outcome because MSACSRNSA is not seasonally adjusted and shows both a slow-moving trend and a recurring annual pattern, alongside serial dependence from one month to the next; regular and seasonal differencing address the trend and seasonality, while autoregressive and moving-average terms capture the remaining short-run dependence (Box, Jenkins, Reinsel, and Ljung 2015). Comparing a univariate time-series benchmark against a specification informed by economic variables also has direct precedent in the housing-forecasting literature: an early comparison of an economic housing-sector model against an ARIMA model for U.S. housing starts found the economic model preferable for near-term forecasts but the ARIMA model more accurate at longer horizons (Puri and Van Lierop 1988), and more recent work continues to evaluate the accuracy of housing-starts forecasts directly (Meyer 2019). Extending this univariate benchmark to a SARIMAX specification is what makes the comparison in this paper worth running: the construction-activity and financing-conditions research discussed above gives a substantive, economically motivated reason to expect HOUST and MORTGAGE30US to carry information beyond the series' own history, and a SARIMAX extension is a direct way to test whether they actually do.

METHODOLOGY AND BASELINE MODEL

A univariate analysis of 757 monthly observations spanning January 1963 to January 2026 was conducted to identify the most appropriate seasonal time series model, following the Box-Jenkins identify-estimate-diagnose cycle (Box, Jenkins, Reinsel, and Ljung 2015). Candidate models with regular and seasonal differencing were compared using information criteria and residual diagnostics on this full historical series, all with a constant term estimated except where noted. Table 1 reports the fit statistics for each candidate.

Candidate Order	AIC	SBC	N
AR(1), seasonal MA(12) [P=1, Q=(12)]	1471.92	1485.76	744
Subset MA at lags 1 and 12 [Q=(1,12)]	1478.94	1492.78	744
AR(1) with subset MA at lags 1 and 12 [P=1, Q=(1,12)]	1473.47	1491.92	744
SARIMA(1,1,0)×$(1,1,1)_{12}$, no constant [P=(1)(12), Q=(12)] — selected	1463.01	1476.85	744

Table 1. Candidate Univariate SARIMA Orders on the Full Historical Series (757 Observations, 744 after Differencing)

The first three candidates include an estimated constant term; the selected order was also estimated without a constant, which further lowers its AIC and SBC relative to the constant-included alternatives

above. Among the four, the selected order SARIMA(1,1,0)×$(1,1,1)_{12}$ achieves the lowest AIC and the lowest SBC, and it is this order, re-estimated on the common 1971–2026 sample described below, that is carried forward as the baseline for the SARIMAX comparisons that follow.

DATA PREPARATION AND STATIONARITY TESTING

Because the SARIMAX comparisons later in this paper require a common sample across MSACSRNSA, HOUST, and MORTGAGE30US, the three series were imported and merged using the SQL procedure before refitting the baseline for comparison. MSACSRNSA (months' supply of new houses, not seasonally adjusted, monthly, January 1963 onward) and HOUST (new privately-owned housing units started, thousands of units at a seasonally adjusted annual rate, monthly, a comparable historical span) are both produced by the U.S. Census Bureau in cooperation with the U.S. Department of Housing and Urban Development as part of the New Residential Sales and New Residential Construction releases, both issued on a monthly release schedule with a roughly three- to four-week reporting lag after each reference month closes. MORTGAGE30US (30-year fixed-rate mortgage average, percent) originates from Freddie Mac's Primary Mortgage Market Survey, reported weekly beginning in April 1971 and released within the same week it is collected. All three series are distributed through the Federal Reserve Economic Data (FRED) database maintained by the Federal Reserve Bank of St. Louis, which is cited here as the data distributor rather than the original producer. Because HOUST and MORTGAGE30US for a given month are not fully known at the time that month's MSACSRNSA figure would be forecast, this release-timing detail underlies the predictor-timing limitation discussed later in this paper.

```
PROC SQL;
   CREATE TABLE my_sarimax_data AS
   SELECT a.DATE, a.MSACSRNSA, b.MORTGAGE30US, c.HOUST
   FROM housing_prep a
   LEFT JOIN houst_prep c ON a.YR = c.YR AND a.MON = c.MON
   LEFT JOIN mortgage_monthly b ON a.YR = b.YR AND a.MON = b.MON
   ORDER BY a.DATE;
QUIT;
```

Because MORTGAGE30US is reported weekly, it is first collapsed to a monthly average, computed as the mean of the weekly survey readings falling within each calendar month, and merged with the other two series on year and month rather than on the exact reporting date, since the weekly and monthly series do not share aligned dates. Across the 663 months in the raw monthly average, this mean is drawn from four weekly readings in 431 months (65.0%) and five in 231 months (34.8%), depending on how many Thursdays, the survey's release day, fall in that month; exactly one month is drawn from only two weekly readings, consistent with a partial month at one boundary of the series rather than a missing-data problem in the interior of the sample. The merged data set is then restricted to the period in which mortgage data are available, April 1971 through January 2026, since HOUST and MSACSRNSA both extend further back than MORTGAGE30US. This common window contains 658 monthly observations; the regular and seasonal differencing used in the SARIMA and SARIMAX models below reduces this to 645 effective observations for estimation. Aside from that one partial-month average, no months in the common window required special handling for missing values.

Each candidate exogenous series was checked for stationarity using the augmented Dickey-Fuller (ADF) test within PROC ARIMA (Dickey and Fuller 1979), evaluated under zero-mean, single-mean, and trend deterministic specifications at lags 0 through 2:

```
PROC ARIMA DATA=mortgage_sarimax_data;
   IDENTIFY VAR=HOUST STATIONARITY=(ADF);
RUN;
QUIT;

PROC ARIMA DATA=mortgage_sarimax_data;
   IDENTIFY VAR=HOUST(1) STATIONARITY=(ADF);
RUN;
QUIT;
```

```
PROC ARIMA DATA=mortgage_sarimax_data;
   IDENTIFY VAR=MORTGAGE30US STATIONARITY=(ADF);
RUN;
QUIT;

PROC ARIMA DATA=mortgage_sarimax_data;
   IDENTIFY VAR=MORTGAGE30US(1) STATIONARITY=(ADF);
RUN;
QUIT;
```

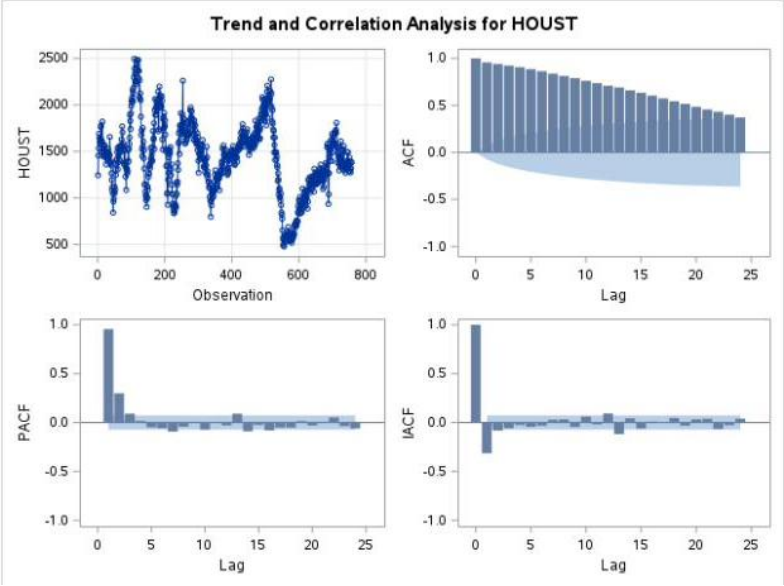


Figure 1. ACF and PACF of Housing Starts (HOUST) in Levels, Showing Strong Persistence and Motivating First Differencing

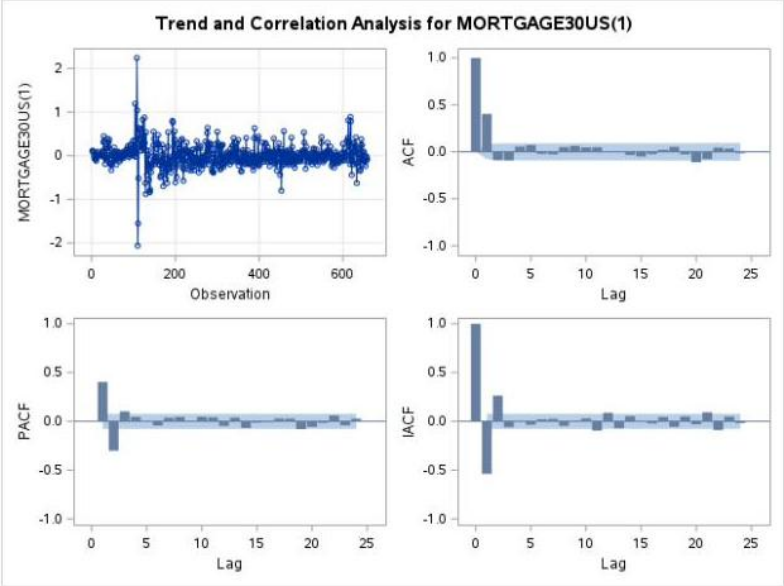


Figure 2. ACF and PACF of Differenced Mortgage Rates, Consistent with Stationarity after First Differencing

For HOUST in levels, the ADF test does not consistently reject a unit root. Under the zero-mean specification, tau statistics range from −1.03 to −1.21 (p = 0.21 to 0.27) across lags 0 through 2; under the single-mean specification, the test rejects at lag 0 (tau = −3.75, p = 0.0038) but not at lag 1 or lag 2 (p = 0.060 and 0.094); and under the trend specification, results are similarly mixed (p = 0.011, 0.184, and 0.275 at lags 0 through 2). This lag- and specification-dependent pattern, together with the slowly decaying autocorrelation function shown in Figure 1, is consistent with strong persistence in the level series rather than a clean, unqualified unit-root finding, and it motivates working with the first-differenced series rather than relying on the level test alone. A direct ADF test of the first-differenced HOUST series rejects the unit-root null decisively across all three deterministic specifications (p < .0001 at every lag examined).

MORTGAGE30US in levels fails to reject a unit root far more uniformly than HOUST does: tau p-values range from 0.39 to 0.47 under the zero-mean specification, from 0.68 to 0.74 under the single-mean specification, and from 0.21 to 0.56 under the trend specification, across lags 0 through 2 in every case. This is consistent with the persistence typical of interest rate series, and unlike HOUST, it does not show a lag-0 rejection under any specification. The first-differenced series, shown in Figure 2, has an autocorrelation function that cuts off quickly, and the ADF test on the differenced series rejects the unit-root null, confirming stationarity. Both HOUST and MORTGAGE30US are therefore first-differenced before entering the SARIMAX models below.

BASELINE MODEL

With the common analytical sample in hand, the SARIMA(1,1,0)×$(1,1,1)_{12}$ order identified above is refit on this restricted sample, so that its fit statistics are directly comparable to the SARIMAX specifications that follow. This baseline model is estimated using the ARIMA procedure:

```
PROC ARIMA DATA=mortgage_sarimax_data;
   IDENTIFY VAR=MSACSRNSA(1,12);
   ESTIMATE P=(1)(12) Q=(12) NOCONSTANT METHOD=ML;
RUN;
QUIT;
```

The IDENTIFY statement differences MSACSRNSA once regularly and once seasonally at lag 12, and the ESTIMATE statement fits an autoregressive term at lag 1, a seasonal autoregressive term at lag 12, and a seasonal moving-average term at lag 12. Written out, the fitted baseline model takes the form:

$$(1 - \phi_1 B)(1 - \Phi B^{12})(1 - B)(1 - B^{12})\, Y_t = (1 - \Theta B^{12})\, e_t$$

where Y_t is the Monthly Supply of New Houses in months, B is the backshift operator, ϕ_1 and Φ are the nonseasonal and seasonal autoregressive coefficients, and Θ is the seasonal moving-average coefficient. e_t is white-noise error. On the 645-observation estimation sample, Maximum Likelihood Estimation gives ϕ_1 = −0.130 (SE = 0.039, 95% CI: −0.206, −0.053, p = 0.0009), Φ = 0.132 (SE = 0.047, 95% CI: 0.040, 0.225, p = 0.0050), and Θ = 0.897 (SE = 0.026, 95% CI: 0.846, 0.949, p < 0.0001), with AIC = 1303.46.

This specification was compared against the alternative univariate orders noted above using Akaike's Information Criterion (AIC) and the Schwarz Bayesian Criterion (SBC). A twelve-month holdout evaluation of this exact model, refit on data through January 2025 and evaluated against the twelve months from February 2025 through January 2026, produced MAE = 0.628 months, RMSE = 0.717 months, and MAPE = 7.02%. This holdout uses a single fixed origin: the model is estimated once on data through January 2025 and forecasts all twelve subsequent months from that origin, rather than being re-estimated or updated month by month as new data arrive; a sequentially updated (rolling-origin) evaluation is a natural robustness check for future work. Table 2 reports the month-by-month actual values, forecasts, 95% prediction intervals, and errors underlying these summary statistics.

Month	Actual	Forecast	L95	U95	APE (%)
Feb 2025	8.700	8.312	7.073	9.550	4.46
Mar 2025	7.900	7.685	6.043	9.327	2.72
Apr 2025	7.600	7.882	5.907	9.858	3.72

May 2025	9.000	8.020	5.761	10.279	10.89
Jun 2025	8.800	8.396	5.884	10.907	4.60
Jul 2025	9.400	8.562	5.822	11.302	8.91
Aug 2025	8.700	8.962	6.011	11.913	3.01
Sep 2025	8.500	9.112	5.964	12.260	7.20
Oct 2025	10.300	9.768	6.435	13.102	5.16
Nov 2025	9.100	10.361	6.852	13.871	13.86
Dec 2025	8.600	9.800	6.123	13.476	13.95
Jan 2026	9.700	9.138	5.302	12.975	5.79

Table 2. Twelve-Month Holdout Detail for the Baseline SARIMA Model (February 2025–January 2026)

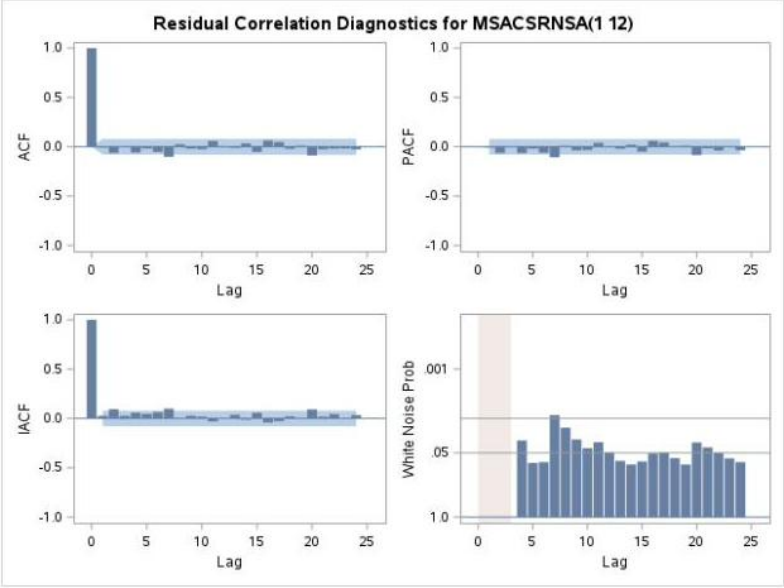


Figure 3. Residual Correlation Diagnostics for the Baseline SARIMA(1,1,0)×(1,1,1)$_{12}$ Model

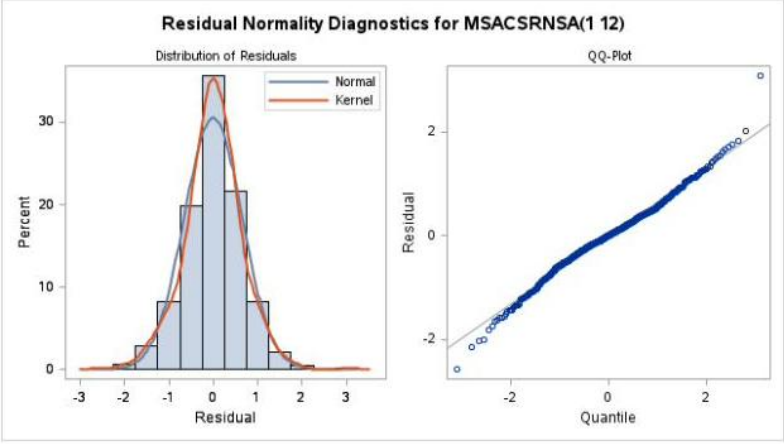


Figure 4. Residual Normality Diagnostics for the Baseline SARIMA(1,1,0)×(1,1,1)$_{12}$ Model

The autocorrelation, partial autocorrelation, and inverse autocorrelation functions in Figure 3 fall within the confidence bounds at nearly every lag, and the white-noise probabilities exceed 0.05 at nearly every lag, with the lag-12 probability at approximately 0.0495, marginally below the conventional 0.05 threshold. This borderline result is best read as a possible signal of residual seasonal dependence rather than as conclusive evidence against the model, and it is one reason the holdout evaluation above is reported alongside the in-sample diagnostics rather than in place of it. Figure 4 shows a residual distribution that is approximately symmetric and bell-shaped, with a QQ-plot that tracks the reference line closely apart from mild departures in the tails. These diagnostics support the baseline model as a starting point before exogenous predictors are added.

SARIMAX EXTENSION AND MODEL COMPARISON

With a validated univariate baseline in hand, this study next tests whether information outside the housing-supply series itself, specifically the pace of new construction and the cost of borrowing, improves model fit. The baseline SARIMA specification is extended into a Seasonal Autoregressive Integrated Moving Average with Exogenous Variables (SARIMAX) model, following SAS's regression-with-ARMA-errors formulation for a simple input series:

$$(1 - B)(1 - B^{12})\, Y_t = \Sigma_i \omega_i (1 - B)\, X_{i,t} + N_t$$

$$(1 - \phi_1 B)(1 - \Phi B^{12})\, N_t = (1 - \Theta B^{12})\, e_t$$

where $X_{i,t}$ is the i-th first-differenced exogenous predictor, ω_i is its regression coefficient, and N_t is the noise process left after removing the regression effect, which itself follows the same seasonal ARMA structure as the baseline model. This two-equation form, rather than folding the AR and MA operators directly onto Y_t alongside the regression term, matches how PROC ARIMA actually estimates a simple INPUT= term: the regression enters first, and the AR and MA operators describe the remaining noise, not Y_t itself. The response MSACSRNSA is both regularly and seasonally differenced, as in the baseline model, while HOUST and MORTGAGE30US enter through a single regular first difference; SAS's reported model specification confirms one period of differencing for each input, versus two (orders 1 and 12) for the response. Each ω_i therefore measures how a one-unit month-to-month change in the exogenous predictor relates to the regularly and seasonally differenced MSACSRNSA series, conditional on the fitted ARMA dynamics, not to an undifferenced or only-once-differenced version of months' supply; because the response is differenced both regularly and seasonally, these coefficients should not be read as direct one-for-one changes in the level of months' supply. HOUST and MORTGAGE30US are first tested one at a time against the baseline, to isolate the contribution of construction activity from the contribution of financial conditions; they are then combined into a single joint model to determine whether their effects are additive or overlapping. Both predictors enter the model as contemporaneous monthly changes. Because a given month's HOUST and MORTGAGE30US values are not necessarily known before that month's MSACSRNSA figure is released, the SARIMAX models below are best interpreted as explanatory, in-sample comparisons of model fit rather than as genuine ahead-of-time forecasting models; producing a true forecast would require lagging the predictors, or forecasting their own future values, which is left for future work.

TESTING EACH PREDICTOR INDIVIDUALLY

The housing-starts specification was fit as follows:

```
PROC ARIMA DATA=mortgage_sarimax_data;
   IDENTIFY VAR=MSACSRNSA(1,12) CROSSCORR=(HOUST(1));
   ESTIMATE P=(1)(12) Q=(12) INPUT=(HOUST) NOCONSTANT METHOD=ML;
RUN;
QUIT;
```

and the mortgage-rate specification as follows:

```
PROC ARIMA DATA=mortgage_sarimax_data;
   IDENTIFY VAR=MSACSRNSA(1,12) CROSSCORR=(MORTGAGE30US(1));
   ESTIMATE P=(1)(12) Q=(12) INPUT=(MORTGAGE30US) NOCONSTANT METHOD=ML;
RUN;
```

```
QUIT;
```

Tested alone, the coefficient on first-differenced HOUST is −0.000486 (SE = 0.000174, 95% CI: −0.000826, −0.000145, p = 0.0051). A one-unit increase in the month-to-month change in HOUST, equivalent to 1,000 additional annualized housing starts relative to the previous month, is thus associated with a 0.000486 decrease in the regularly and seasonally differenced MSACSRNSA response, conditional on the fitted ARMA dynamics. This specification lowers the AIC to 1297.98; its residual diagnostics show a significant lag-12 autocorrelation (Pr>ChiSq = 0.0380). For MORTGAGE30US, the coefficient on the first-differenced series is 0.197 (SE = 0.047, 95% CI: 0.104, 0.289, p < 0.0001). A one-percentage-point increase in the month-to-month change in the 30-year mortgage rate is thus associated with a 0.197 increase in that same differenced MSACSRNSA response, conditional on the fitted ARMA dynamics. This specification produces an AIC of 1288.13 and an SBC of 1306.00, with clean residuals at every lag checked (Pr>ChiSq ≥ 0.09 throughout). Because the response is differenced regularly and seasonally, these coefficients should not be interpreted as direct one-for-one changes in the level of months' supply. These are conditional associations within the fitted time series model, not estimates of a causal effect. Residual diagnostics are therefore strongest for the mortgage-only and joint specifications, with isolated borderline or significant lag-12 results for the baseline and housing-starts-only models. Table 3 summarizes both specifications alongside the univariate baseline.

COMBINING BOTH PREDICTORS: THE FINAL MULTIVARIATE MODEL

Because each predictor improves on the baseline individually, they are combined into a single multivariate SARIMAX model with HOUST and MORTGAGE30US together:

```
PROC ARIMA DATA=mortgage_sarimax_data;
   IDENTIFY VAR=MSACSRNSA(1,12) CROSSCORR=(HOUST(1) MORTGAGE30US(1));
   ESTIMATE P=(1)(12) Q=(12) INPUT=(HOUST MORTGAGE30US) NOCONSTANT
METHOD=ML;
RUN;
QUIT;
```

Table 3 compares all four models, the univariate baseline, each single-predictor SARIMAX specification, and the final joint multivariate model, on AIC, SBC, and the significance of the exogenous coefficient(s).

Model	AIC	SBC	Key Parameter	Estimate	p-value
Baseline SARIMA(1,1,0)×$(1,1,1)_{12}$	1303.46	1316.87	—	—	—
SARIMAX with Housing Starts	1297.98	1315.85	HOUST	−0.000486	0.0051
SARIMAX with Mortgage Rates	1288.13	1306.00	MORTGAGE30US	0.197	< 0.0001
Final Multivariate SARIMAX (HOUST + MORTGAGE30US)	1286.59	1308.94	HOUST MORTGAGE30US	−0.000341 0.178	0.0569 0.0002

Table 3. Comparison of SARIMA and SARIMAX Models

The joint model achieves the lowest AIC of any specification tested (1286.59), with ϕ_1 = −0.136, Φ = 0.137, and Θ = 0.895. It does not, however, have the lowest SBC: the mortgage-only model's SBC of 1306.00 is actually lower than the joint model's 1308.94, a difference of about 2.9 points, while the joint model's AIC advantage over mortgage-only is only about 1.5 points (1288.13 versus 1286.59). AIC and SBC therefore disagree, with SBC's heavier penalty for the extra parameter favoring the more parsimonious mortgage-only specification. In the joint model, the mortgage-rate coefficient remains strongly significant (ω = 0.178, SE = 0.048, 95% CI: 0.083, 0.273, p = 0.0002), essentially unchanged from the single-predictor model, but the housing-starts coefficient attenuates sharply, from ω = −0.000486 (95% CI: −0.000826, −0.000145) when tested alone to ω = −0.000341 (SE = 0.000179, 95% CI: −0.000692, 0.000010) once mortgage rate is included, a confidence interval that now spans zero,

consistent with its p = 0.0569. At the conventional α = 0.05 threshold, HOUST is no longer statistically significant once MORTGAGE30US is accounted for.

A Pearson correlation between the two first-differenced regressors reveals a modest, though statistically significant, negative association (r = −0.130, p = 0.0008, n = 657); with this many observations, even a weak correlation reaches significance, so the coefficient alone should not be read as evidence of strong collinearity. It is consistent with the two series sharing some common variation, but the correlation by itself does not establish that collinearity is what drives HOUST's attenuation; the coefficient shift and the information-criteria comparison above are the more direct evidence that housing starts contribute relatively little incremental information once mortgage rates are already in the model. Substantively, this pattern is consistent with a mechanism in which higher borrowing costs are associated with cooler buyer demand, slower sales, and greater months' supply; mortgage rates appear to carry much of the incremental signal that housing starts capture on their own. Because HOUST measures the pace of new construction starts rather than completions, this reflects construction-cycle information rather than a direct completed-homes mechanism. The comparisons above describe in-sample fit; a conditional holdout forecast comparison, reported later in this section, is needed to establish whether these predictors also improve forecast accuracy and not fit alone.

Figures 5 and 6 present the residual diagnostics for the mortgage-only SARIMAX model, and Figures 7 and 8 present the same diagnostics for the final joint multivariate model.

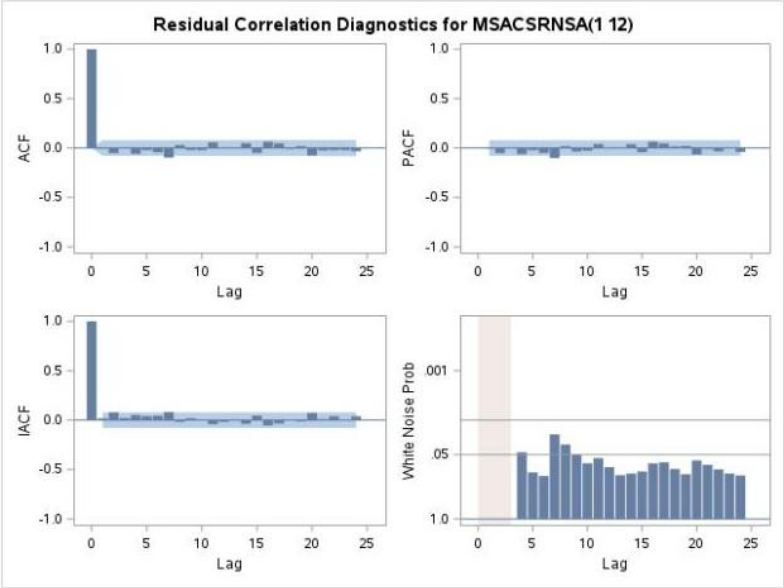


Figure 5. Residual Correlation Diagnostics for the SARIMAX Model with Differenced Mortgage Rates Only

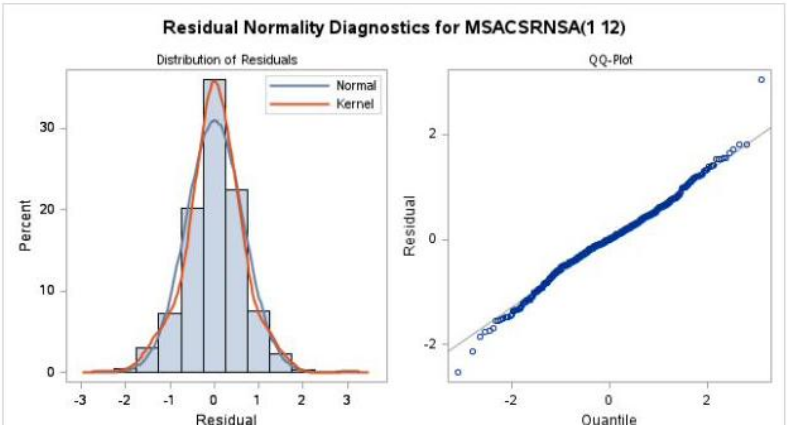


Figure 6. Residual Normality Diagnostics for the SARIMAX Model with Differenced Mortgage Rates Only

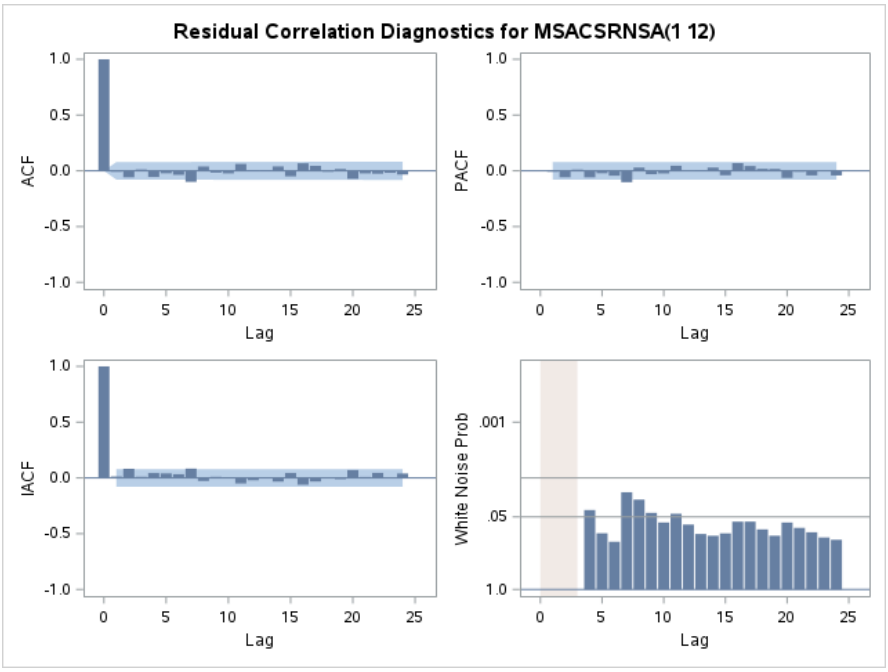


Figure 7. Residual Correlation Diagnostics for the Final Multivariate SARIMAX Model (HOUST and MORTGAGE30US)

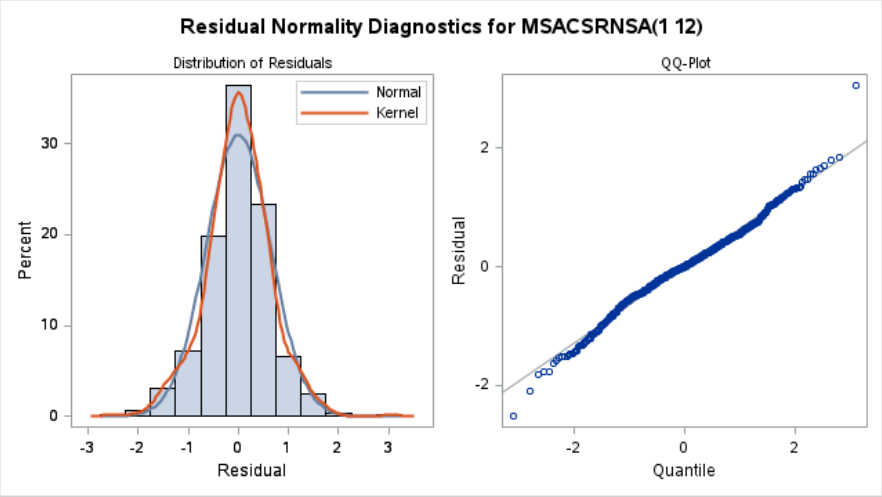


Figure 8. Residual Normality Diagnostics for the Final Multivariate SARIMAX Model (HOUST and MORTGAGE30US)

Across all four figures, the autocorrelation and partial autocorrelation functions remain within their confidence bounds, and the white-noise probabilities for the mortgage-only and joint models stay above 0.05 at every lag checked, indicating no significant residual structure in either specification. The residual distributions and QQ-plots in Figures 6 and 8 are both close to normal, with only mild departures in the tails. Adding the second regressor therefore comes at no cost to residual adequacy and improves the AIC, though, as noted above, it does not improve the SBC relative to the mortgage-only model.

CONDITIONAL HOLDOUT FORECAST COMPARISON

The comparisons above rely on AIC, SBC, coefficients, and residual diagnostics, which describe in-sample model fit rather than forecast accuracy. To address this directly, each SARIMAX specification was evaluated on the same twelve-month holdout window used for the baseline (February 2025–January 2026), alongside a seasonal-naive benchmark that simply forecasts each month as the actual value from twelve months earlier. Because a SARIMAX forecast requires future values of its exogenous predictors, these holdout forecasts use the realized (actual, historical) HOUST and MORTGAGE30US values for the holdout months rather than values that were genuinely known in advance; this measures conditional forecast accuracy given perfect foreknowledge of the exogenous inputs, not a true ex-ante forecast, consistent with the predictor-timing limitation noted earlier in this paper.

Model	MAE	RMSE	MAPE (%)
Baseline SARIMA (univariate)	0.628	0.717	7.02
SARIMAX: HOUST only	0.614	0.705	6.91
SARIMAX: Mortgage rate only	0.584	0.766	6.64
SARIMAX: Joint (HOUST + Mortgage)	0.590	0.767	6.74
Seasonal-naive benchmark	0.817	0.953	9.20

Table 4. Same-Window Twelve-Month Holdout Comparison (February 2025–January 2026)

All four fitted models produced lower MAE, RMSE, and MAPE than the seasonal-naive benchmark (MAE = 0.817, RMSE = 0.953, MAPE = 9.20%) over this twelve-month window. Relative to the univariate baseline, the picture is mixed rather than uniform. The HOUST-only specification improves on the baseline on all three metrics (MAE 0.614 vs. 0.628, RMSE 0.705 vs. 0.717, MAPE 6.91% vs. 7.02%). The mortgage-only and joint specifications improve on MAE and MAPE (0.584/6.64% and 0.590/6.74%, respectively, both below the baseline's 0.628/7.02%) but have higher RMSE than the baseline (0.766 and 0.767 versus 0.717), indicating that while their average errors are smaller, they also produced at least

one larger miss during this twelve-month window that RMSE, which squares errors before averaging, penalizes more heavily than MAE does. Under the caveat that these forecasts assume perfect foreknowledge of the exogenous inputs, the evidence therefore supports a modest, metric-dependent forecast improvement from the exogenous predictors rather than a uniform one: HOUST's inclusion helps by every measure considered here, while mortgage rate's inclusion helps on typical-error measures but not on the measure most sensitive to large misses.

CONCLUSION

This paper applied a two-stage Box-Jenkins approach to forecasting the Monthly Supply of New Houses in the United States. In the first stage, a SARIMA$(1,1,0)\times(1,1,1)_{12}$ model without a constant term was identified as a robust univariate baseline, supported by information criteria and residual diagnostics, and confirmed via a twelve-month holdout evaluation of this exact specification (MAE = 0.628, RMSE = 0.717, MAPE = 7.02%). In the second stage, this baseline was extended into a multivariate SARIMAX framework to test whether new housing starts and 30-year mortgage rates, each first-differenced to satisfy stationarity, improve on the univariate model, first individually and then jointly, in both in-sample fit and conditional holdout forecast performance.

Both predictors were statistically significant on their own, with the mortgage-rate specification producing the larger improvement in AIC and stronger statistical evidence (AIC = 1288.13 for mortgage versus 1297.98 for HOUST). Combined into a single multivariate model, the two predictors achieved the lowest AIC of any specification tested (1286.59), but not the lowest SBC: the mortgage-only model's SBC (1306.00) is actually lower than the joint model's (1308.94). AIC and SBC therefore point in different directions, and the mortgage-rate coefficient remained significant ($p = 0.0002$) while the housing-starts coefficient weakened to marginal significance ($p = 0.0569$), consistent with, though not fully explained by, a modest negative correlation between the two differenced series ($r = -0.130$, $p = 0.0008$). Once financing conditions are already represented in the model, construction activity contributes comparatively little independent explanatory power, and the more parsimonious mortgage-only specification is at least as defensible a choice as the joint model.

A same-window holdout comparison, using the realized HOUST and MORTGAGE30US values for the holdout months, showed that all three SARIMAX specifications produced lower MAE, RMSE, and MAPE than a seasonal-naive benchmark over this conditional holdout window. Against the univariate baseline specifically, the housing-starts specification improved forecast accuracy on every metric considered, while the mortgage-only and joint specifications improved typical-error measures (MAE, MAPE) but not root-mean-squared error, which is more sensitive to occasional large misses. The forecast-accuracy evidence therefore only partly mirrors the in-sample fit results: mortgage rate is the stronger predictor by AIC and coefficient significance, but HOUST performs more consistently across the three forecast-error metrics in this particular twelve-month holdout window.

These results suggest that borrowing costs, more than construction activity itself, are associated with short-run swings in new-home months' supply, and that some of what housing starts appears to explain on its own overlaps with the cost of borrowing. For forecasters and policymakers, this still implies that models of housing-supply volatility built only on construction data risk understating the role of financing conditions, though the mixed RMSE result above is a reminder that no single specification dominates on every measure.

Because this study examined only two exogenous predictors, a natural extension would test additional candidates, such as building permits, existing-home inventory, or measures of buyer demand, to determine whether they carry information distinct from both construction activity and borrowing costs. Because the holdout comparison above assumes perfect foreknowledge of HOUST and MORTGAGE30US, a further useful extension would forecast the exogenous predictors themselves, or use genuinely lagged inputs, to obtain a true ex-ante forecast rather than a conditional one; re-validating the model with a rolling-origin holdout and as new data become available would further test how well these relationships hold outside the sample and window studied here.

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CONTACT INFORMATION

Your comments and questions are valued and encouraged. Contact the authors at:

Matilda Kumi
Department of Mathematics and Statistics, East Tennessee State University
kumim1@etsu.edu

Mostafa Zahed
Department of Mathematics and Statistics, East Tennessee State University
zahedm@etsu.edu